Multiplying and Dividing in Scientific Notation

Question Bank - Additional 4.02

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(3 \times 10^2)(3 \times 10^6)$

2. $\frac{6.3 \times 10^7}{2.1 \times 10^3}$

3. $(4 \times 10^5)(8 \times 10^6)$

4. $\frac{2.4 \times 10^4}{8 \times 10^7}$

5. $(2.5 \times 10^{-5})(6 \times 10^{-4})$

6. $\frac{9.9\times10^{-3}}{3.3\times10^{-8}}$

Real quantities.

7. The Amazon River discharges about 5.5×10^{7} gallons of water every second.

a. About how many gallons does it discharge in one hour, which is 3.6×10^{3} seconds?

b. A second river discharges about 1.1×10^{7} gallons per second. How many times larger is the Amazon's rate?

8. A factory produces about 6.4×10^{5} bottles per day.

a. How many bottles does it produce in 3.5×10^{2} days?

b. A second factory produced 7.2×10^{8} bottles in 1.2×10^{3} days. How many bottles per day is that?

Working backwards.

9. Find the missing number in each equation.

a. $(4\times 10^{n})(2\times 10^{3}) = 8\times 10^{10}$

b. $\frac{7.2\times 10^{6}}{n\times 10^{2}} = 2.4\times 10^{4}$

c. $(2.5\times 10^{3})(8\times 10^{5}) =$ _____

Reading someone else's work.

10. Dana divides $\frac{4.8\times 10^{11}}{1.2\times 10^{4}}$ and writes 3.6×10^{7}. Name her mistake and give the correct quotient.

11. Which is the value of $\frac{9.6\times10^{-3}}{3.2\times10^{5}}$?

- ○ A. 3×10^{-8}
- ○ B. 3×10^{8}
- ○ C. 6.4×10^{-8}
- ○ D. 3×10^{2}

12. Select ALL of the expressions whose value is 8×10^{6} .

- ☐ A. $(1.6\times10^{3})(5\times10^{3})$
- ☐ B. $\frac{1.6\times10^{10}}{2\times10^{3}}$
- ☐ C. $(4\times10^{2})(2\times10^{8})$
- ☐ D. $\frac{4\times10^{8}}{5\times10^{1}}$
- ☐ E. $(4\times10^{2})(2\times10^{4})$
- ☐ F. $\frac{1.6\times10^{10}}{2\times10^{4}}$